

List of Resources

Module 1

Check out the cost here

<https://livechatai.com/gpt-4-5-pricing-calculator>

Create a simple chatbot for a use case of your choice

<https://app.livechatai.com/>

LLM Prompting with LangChain

https://colab.research.google.com/drive/16yHJTuvmVg2EZd89o-RMNPUTuJKq_jX_?usp=sharing

Problem Statement

Build a **Context-Aware Q&A Chatbot** Using OpenAI or Gemini API and **LangChain**. **Objective**

Create a Python chatbot that answers user questions while maintaining the context of the conversation. The chatbot should:

- Use OpenAI API or Google's Gemini API via LangChain (v0.3.26)
- Insert previous chat history dynamically into prompts
- Distinguish between system, human, and AI messages
- Allow the user to ask at least three consecutive questions, with the bot's answers reflecting the ongoing context

Prompt Structure

- a. Begin every conversation with a system message:
"You are a helpful assistant for technology-related questions."
- b. Dynamically insert all previous chat messages (both user and bot) before each new question.
- c. Add the latest user question as a human message.

Conversation Flow

- d. The user should be able to enter at least three questions in sequence.
- e. The bot's response must consider the entire conversation history for each reply.

Implementation Details

- f. Use LangChain's `ChatPromptTemplate` and `MessagesPlaceholder`.
- g. Do not hard-code the API key in the script; read it from an environment variable for security.

Module 2

Advanced Prompting Patterns

https://colab.research.google.com/drive/1ZmikhPmoLdb_Em_N6sMMdwMFaTmHhmvY?usp=sharing

Problem Statement (Group Activity) **Advanced**

Driver Safety Alert System

Problem Context

As a Senior AI Engineer at **TechDrive Automotive**, you're tasked with developing an intelligent driver safety alert system that generates contextually appropriate warning messages for various driving scenarios. The system must provide clear, actionable, and appropriately urgent alerts while avoiding generic or potentially confusing messages.

Business Requirements

Your GenAI system needs to:

- Generate personalized safety alerts based on driving conditions, vehicle status, and

driver behavior

- Adapt message tone and urgency based on severity levels
- Provide specific, actionable guidance rather than generic warnings
- Consider driver demographics and experience levels
- Avoid alerts that could cause panic or distraction

Details here

<https://drive.google.com/file/d/1anISpbaxVFEwj9tWTUsLZLxyRkXUUNTe/view?usp=sharing>

Module 3:

Using Selectors | COT | Self Consistency | ToT Prompts

<https://colab.research.google.com/drive/1PZCFV7U9Mew8wewL8URPJ6oB1sBZQhvh?usp=sharing>

Module 4:

Building conversational applications with Memory

<https://colab.research.google.com/drive/1RsT7wBe3eTumEHh7iwG4-zpJyxnQfaYn?usp=sharing>

AI Financial Advisor Challenge: Advanced Memory Management & Prompting

Problem Statement

You're tasked with building **"WealthWise AI"** - an intelligent financial advisory chatbot that provides personalized investment advice, tracks user portfolios, and maintains context across multiple conversation sessions. The system must handle complex financial scenarios while ensuring responsible AI practices.

For details - Refer

[**AI Financial Advisor Challenge**](#)

Module 7

Practical Try-Outs - Programmatic prompt evaluation with LangChain

<https://colab.research.google.com/drive/1HxWpSzYyOuUFvborMgk4TBtFGS34ey9Z?usp=sharing>

Evaluation using **3rd Party Library** - LangSmith Hub (30 mins)

<https://colab.research.google.com/drive/1-N6GbpAefC84bagpBeu5SKj7u-y8l3zl?usp=sharing>

Module 8:

Practical Try-out - Design Patterns

<https://colab.research.google.com/drive/1Re6XLaLnk5C8QYp1xVTjCbgJCDmjOixP?usp=sharing>

Prompt Reusability | Design Patterns (20 mins)

https://colab.research.google.com/drive/1_oDHV2RUU-Bqzk1iOoUq5WpLc6PuQjlp?usp=sharing

PROJECT GUIDE DOCUMENTATION

https://drive.google.com/file/d/1uMzbxWkFjy5_wN9r-mC85IRb2wv9gsd7/view?usp=sharing